

How do spatial indicators shape emotion during urban walking?

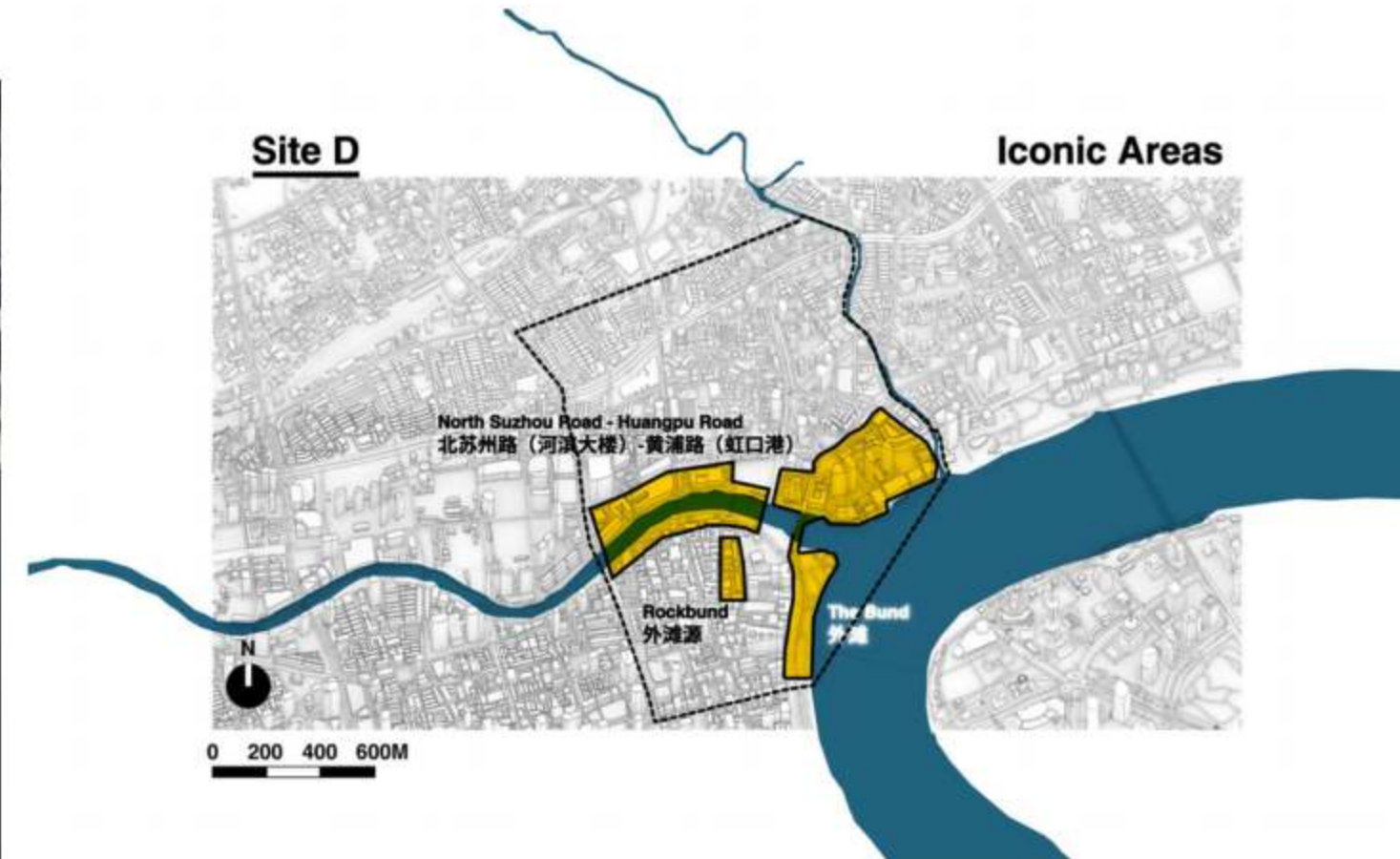
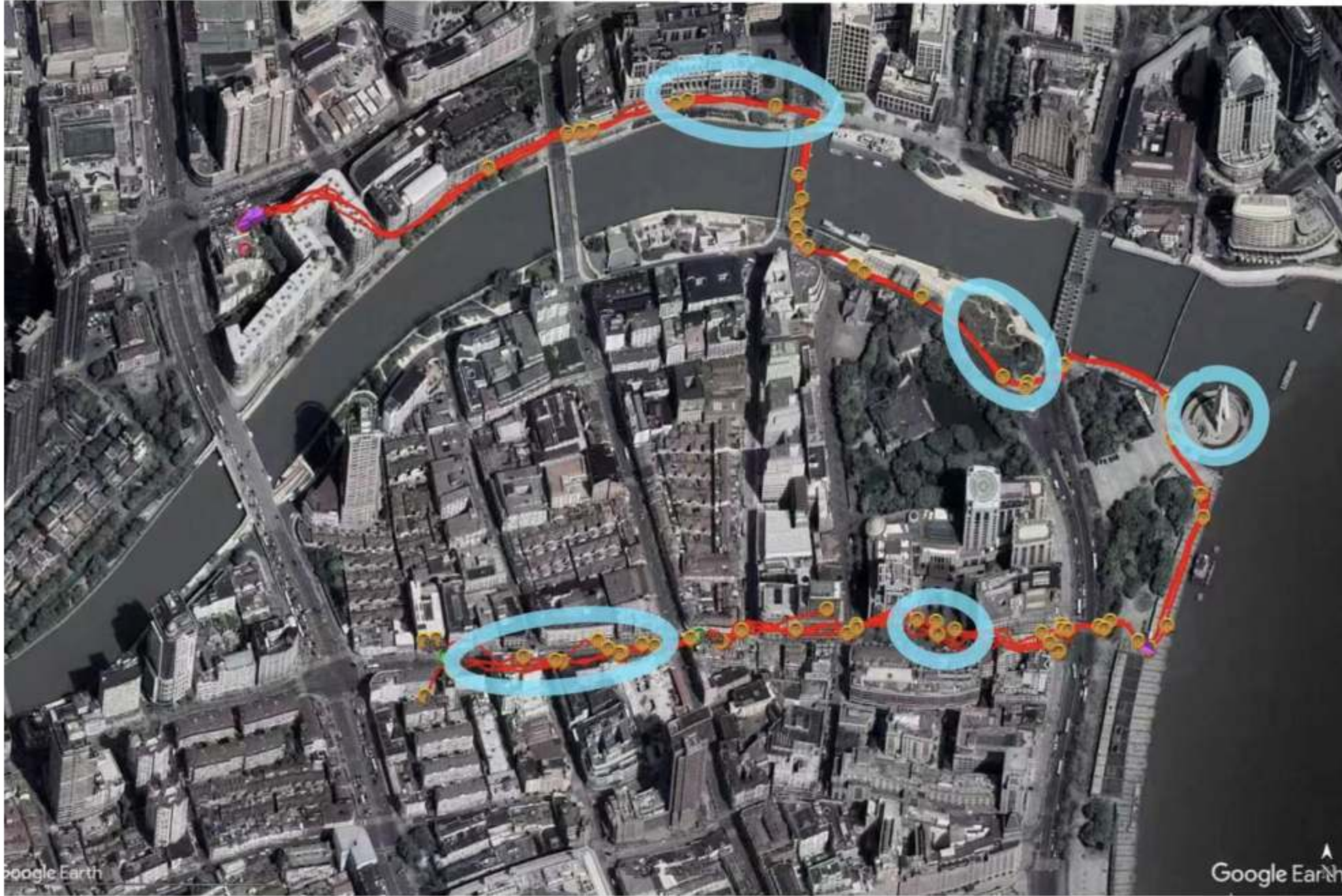
Environmental, enclosure and human-scale cues — including Personal Distance

D Group · 3D



Selected Site with Trajectory Map

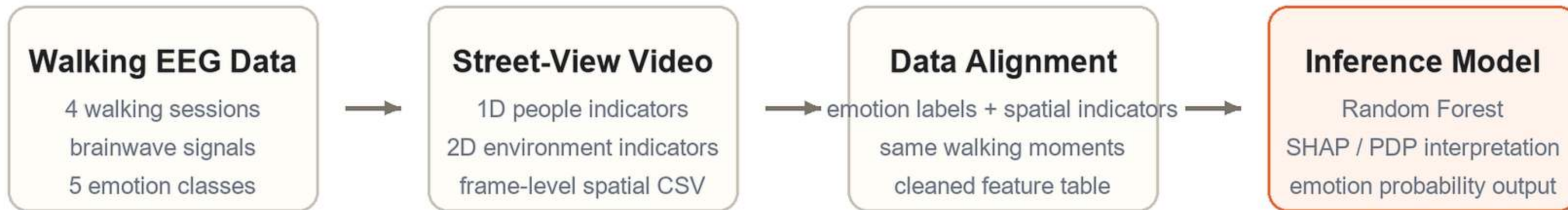
Walking Route and Key Observation Nodes



Mainly covers the **Beijing East Road – The Bund** walking route, including riverside spaces, historic streets, intersections, and landmark nodes.

Research Workflow

From walking EEG and street-view spatial indicators to interpretable emotion prediction



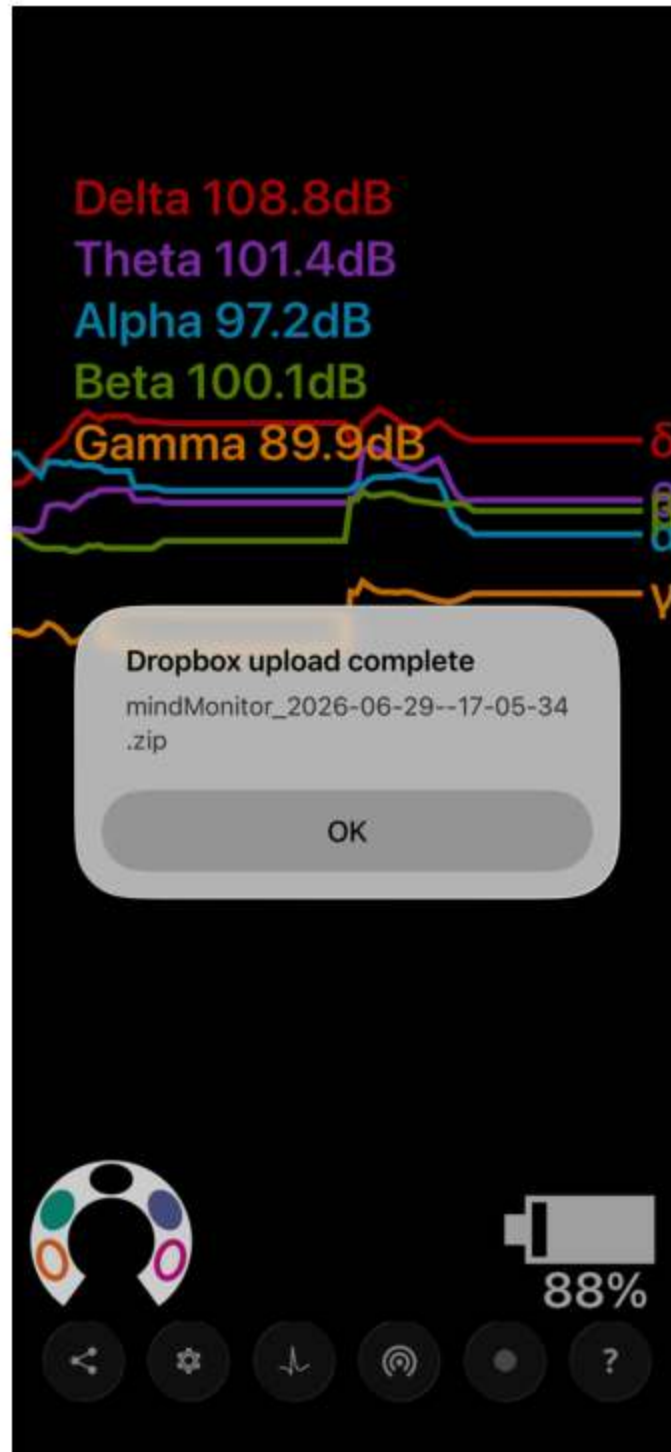
Main research question

How much does Personal Distance contribute to emotion changes during urban walking, compared with other spatial indicators?

Output target: emotion prediction in Rhino scenes of real urban environments

Data Collection

EEG Data + GPS Tracks + Simulated Human Vision



+



+



All have the same timestamp

Image Segmentation

Selected env indexes + Nearest_personal_distance

Classification	Indicator name	Explanation	Calculation Method	Data Source
1D : Crowd Density	Number of Visible Pedestrians	Counts the total number of pedestrians visible within the street scene	YOLO pedestrian detection (or directly Head count)	On-site Photo
1D : Crowd Density	Pedestrian Pixel Ratio	Measures the proportion of the image area occupied by pedestrians	Semantic segmentation	On-site Photo
2D : Spatial Enclosure	Building Pixel Ratio	Measures the proportion of the image area occupied by building façades	Semantic segmentation	On-site Photo
2D : Spatial Enclosure	Sky View Ratio	Measures the degree of spatial openness through the visible proportion of sky	Semantic segmentation	On-site Photo
2D : Spatial Enclosure	Street Height-to-Width Ratio	Describes the vertical enclosure of the street based on building height and street width	Measurement and Estimation	On-site measurement Height estimation

Image Segmentation

Selected env indexes + nearest_personal_distance

Classification	Indicator name	Explanation	Calculation Method	Data Source
2D : Spatial Enclosure	Green View Index	Measures the proportion of visible greenery within the street scene	Semantic segmentation	On-site Photo
2D : Spatial Enclosure	Sidewalk Width	Indicates the amount of lateral space available for pedestrian movement and avoidance	Measurement and Estimation	On-stie measurement
2D : Spatial Enclosure	Visual Permeability	Measures how easily views can extend through or beyond the street space on 1st floor	Semantic segmentation	On-site Photo
3D : Perception	EEG Data	Simply measure on site	On-site measurement	On-site measurement
3D : Perception	Personal Distance	Represents the interpersonal distance at which participants begin to feel comfortable or stressed.	Estimation	On-site Photo Estimation

Image Segmentation

Crowd Density



HeadCount: 11
Nearest Person Distance:~1.59m



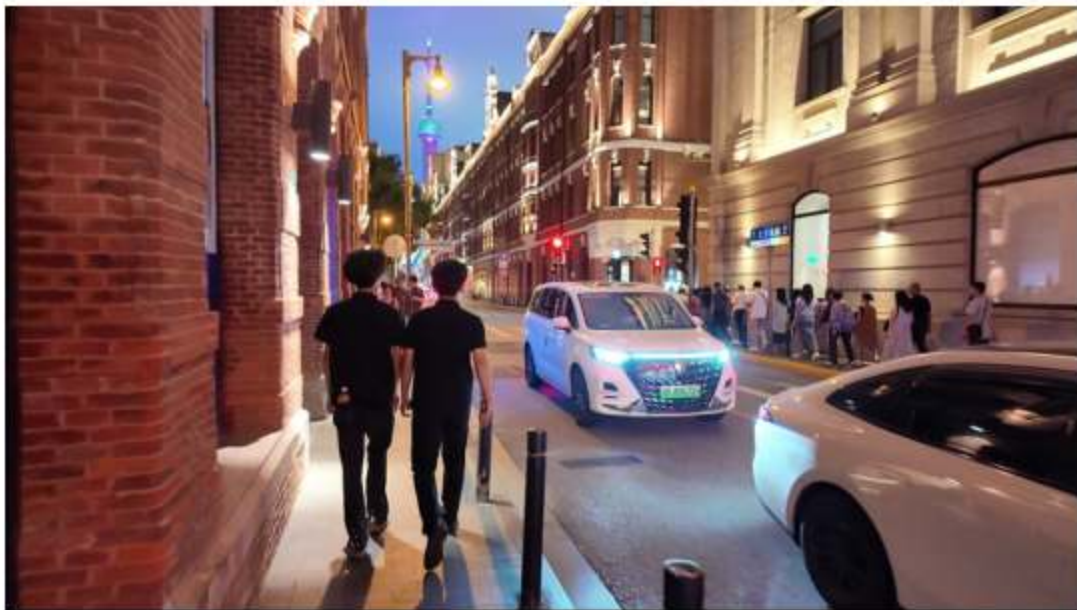
HeadCount:6
Nearest Person Distance:~1.11m

Image Segmentation

Spatial Enclosure



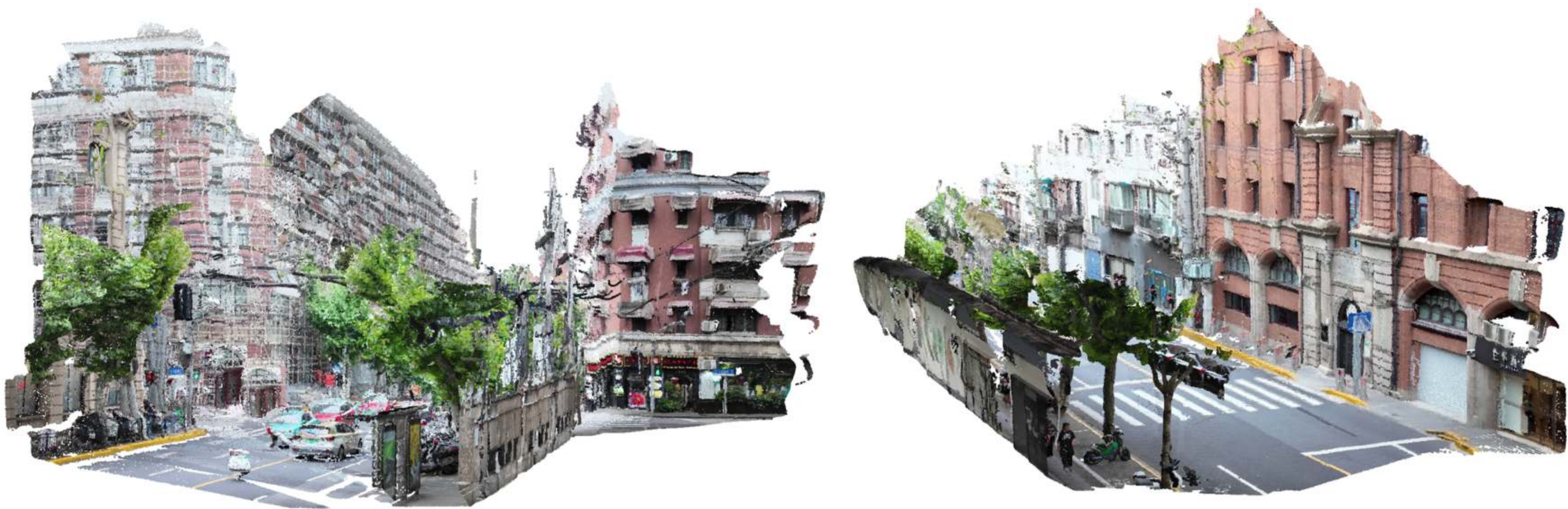
semantic_sky	semantic_building	semantic_person	semantic_vegetation	semantic_vehicle	Height_to_Width	Sidewalk_width	WWR
5.11	61.1	13.68	6.7	2.08	6.584615385	5	0.141641



semantic_sky	semantic_building	semantic_person	semantic_vegetation	semantic_vehicle	Height_to_Width	Sidewalk_width	WWR
2.97	57.05	8.63	0.8	15.61	3.696666667	2	0.097838

3D Reconstruction based on 2D Images

Simulate the real-world personal distance



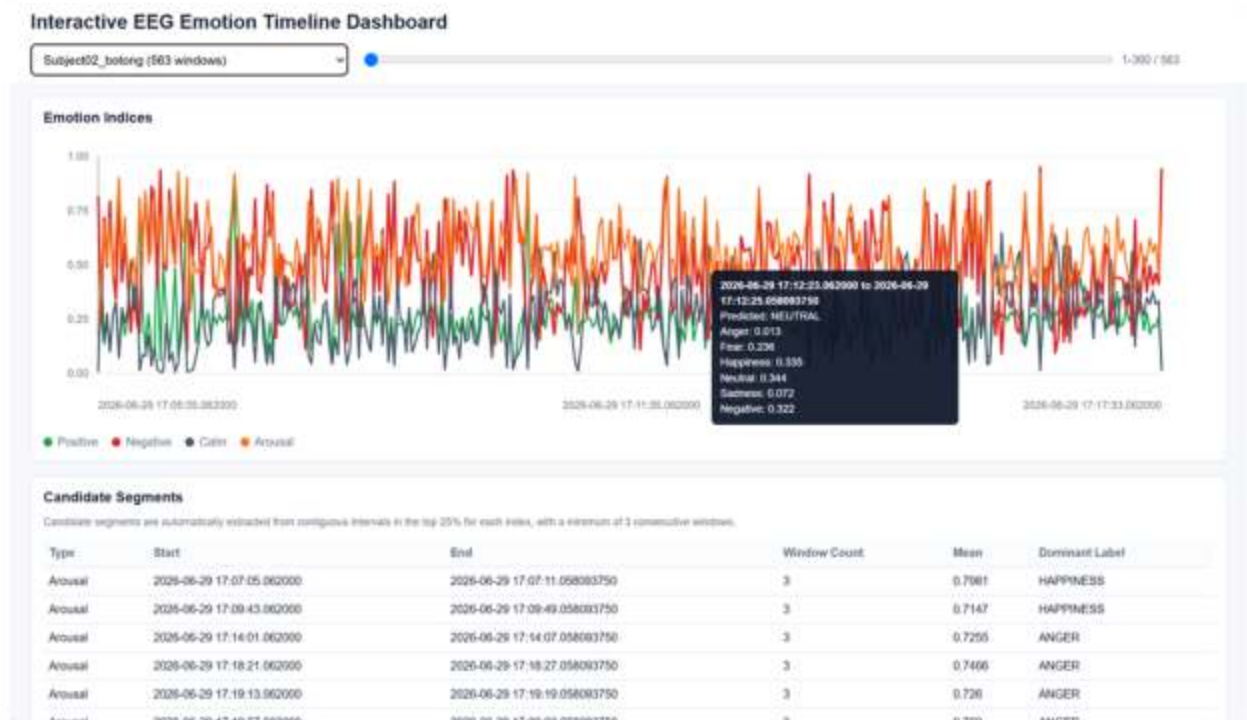
EEG Data

Brain Waves

Subject_01



Subject_02



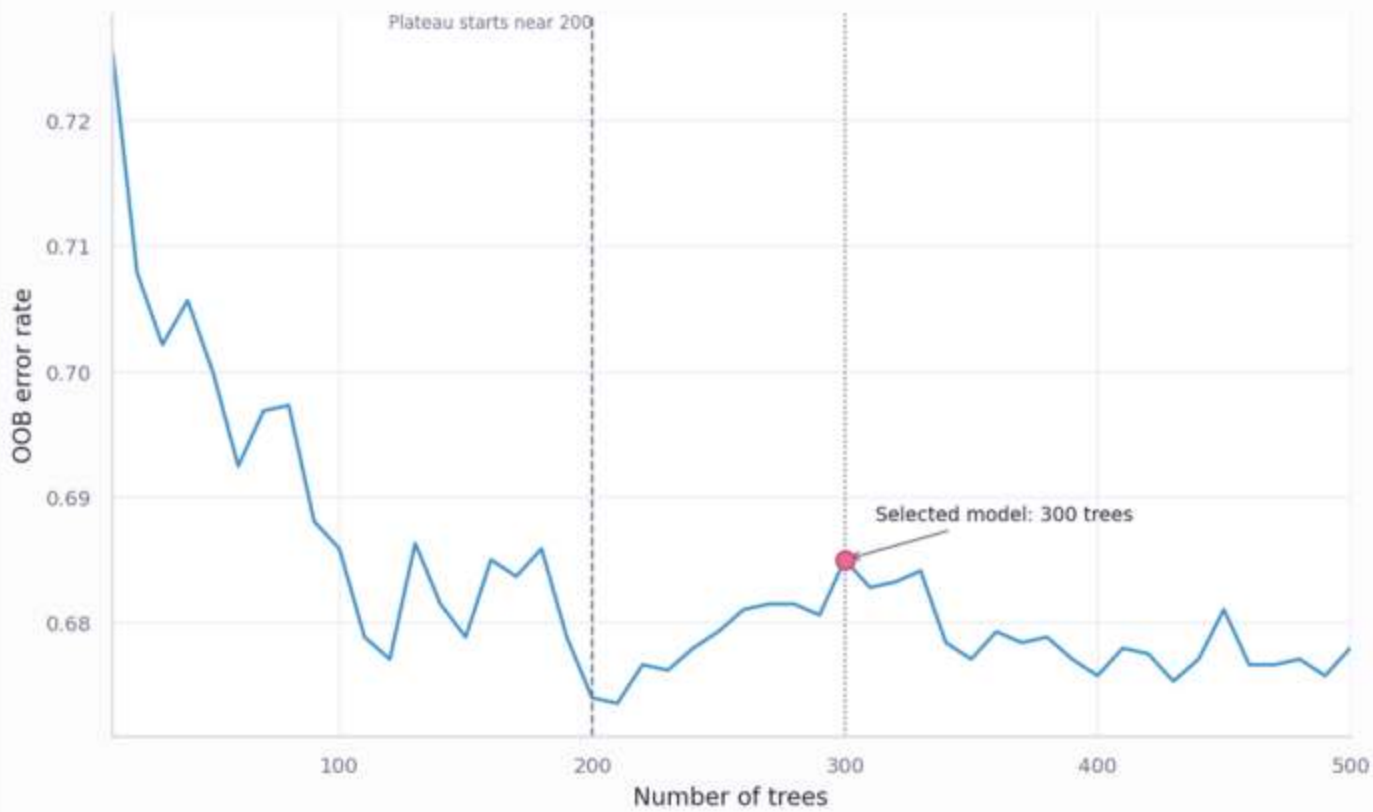
Emotions Categories: Anger Fear Happiness Netural Sadness

RF:Exploring Streetscape Contributions to Emotion Prediction

Model Stability and Test-Set Performance

Random Forest out-of-bag error curve

The curve plateaus after about 200 trees; 300 lies inside the stable region



Random Forest test-set classification report

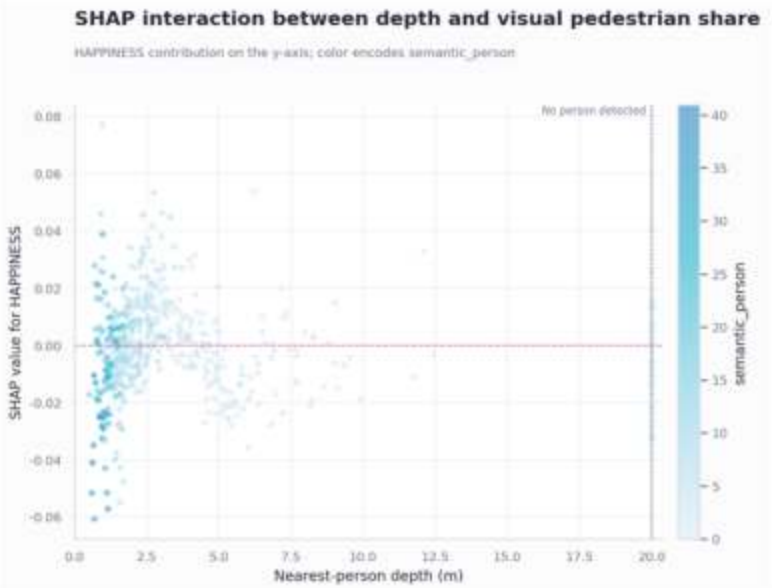
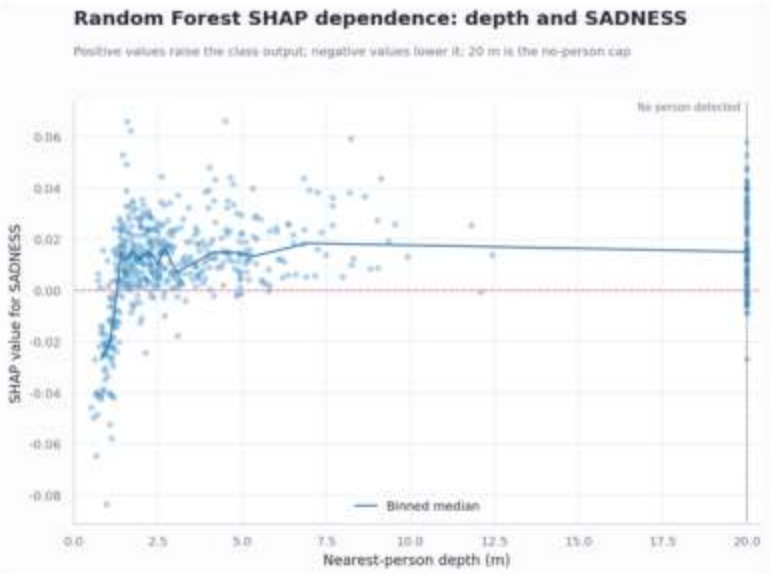
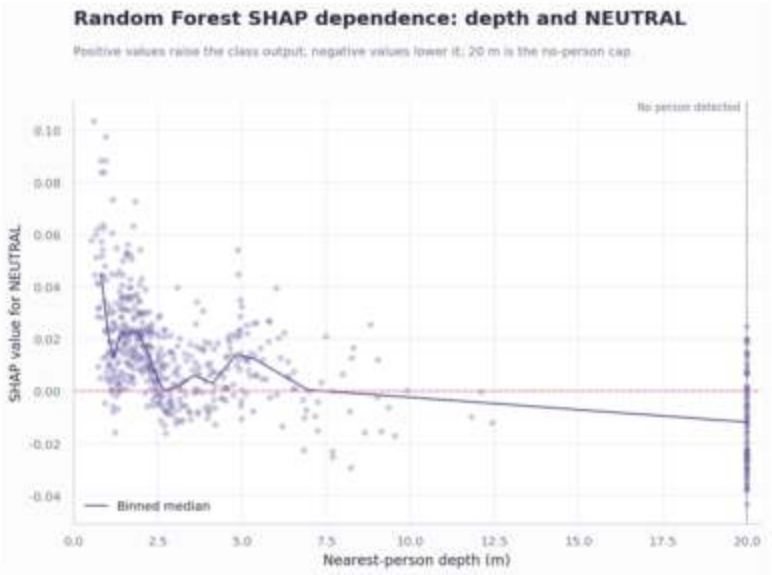
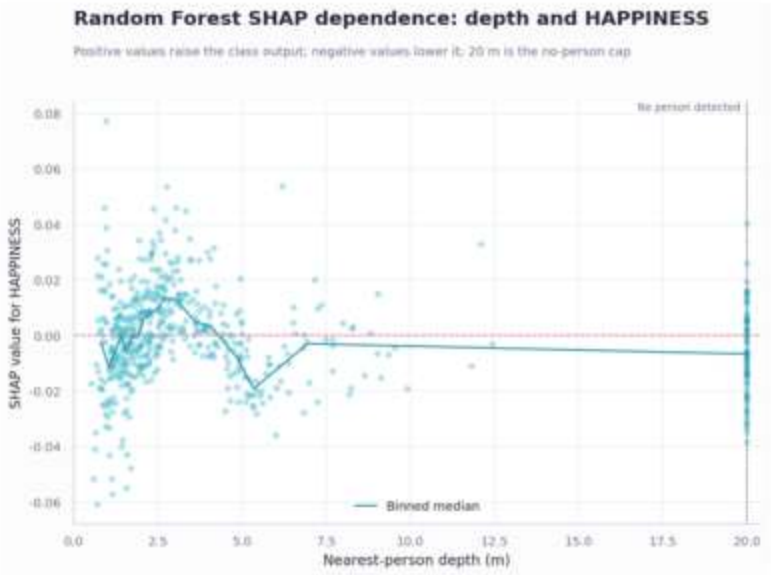
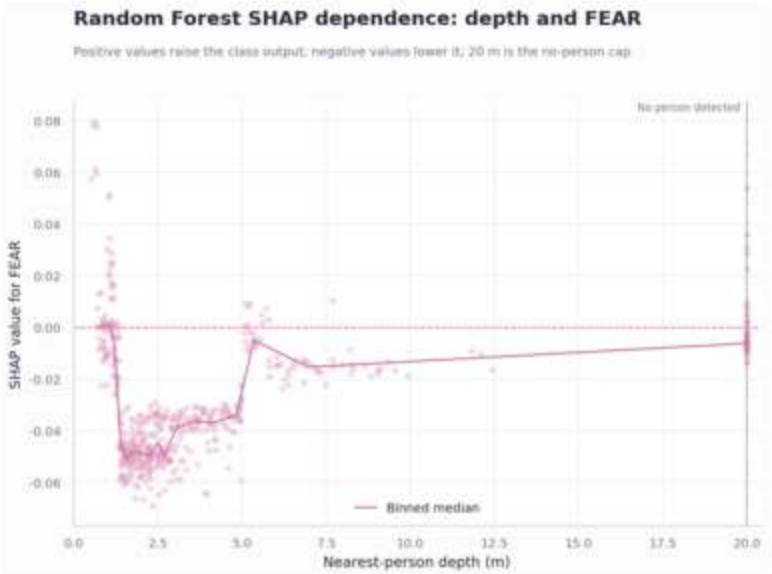
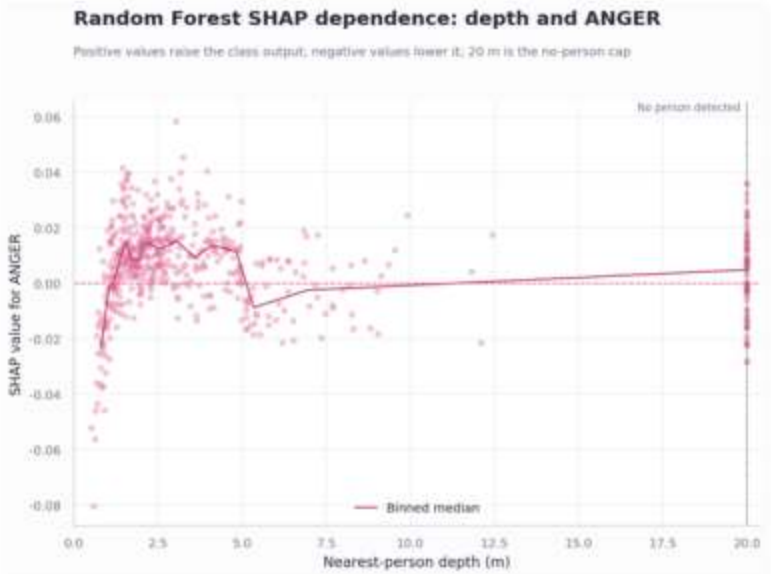
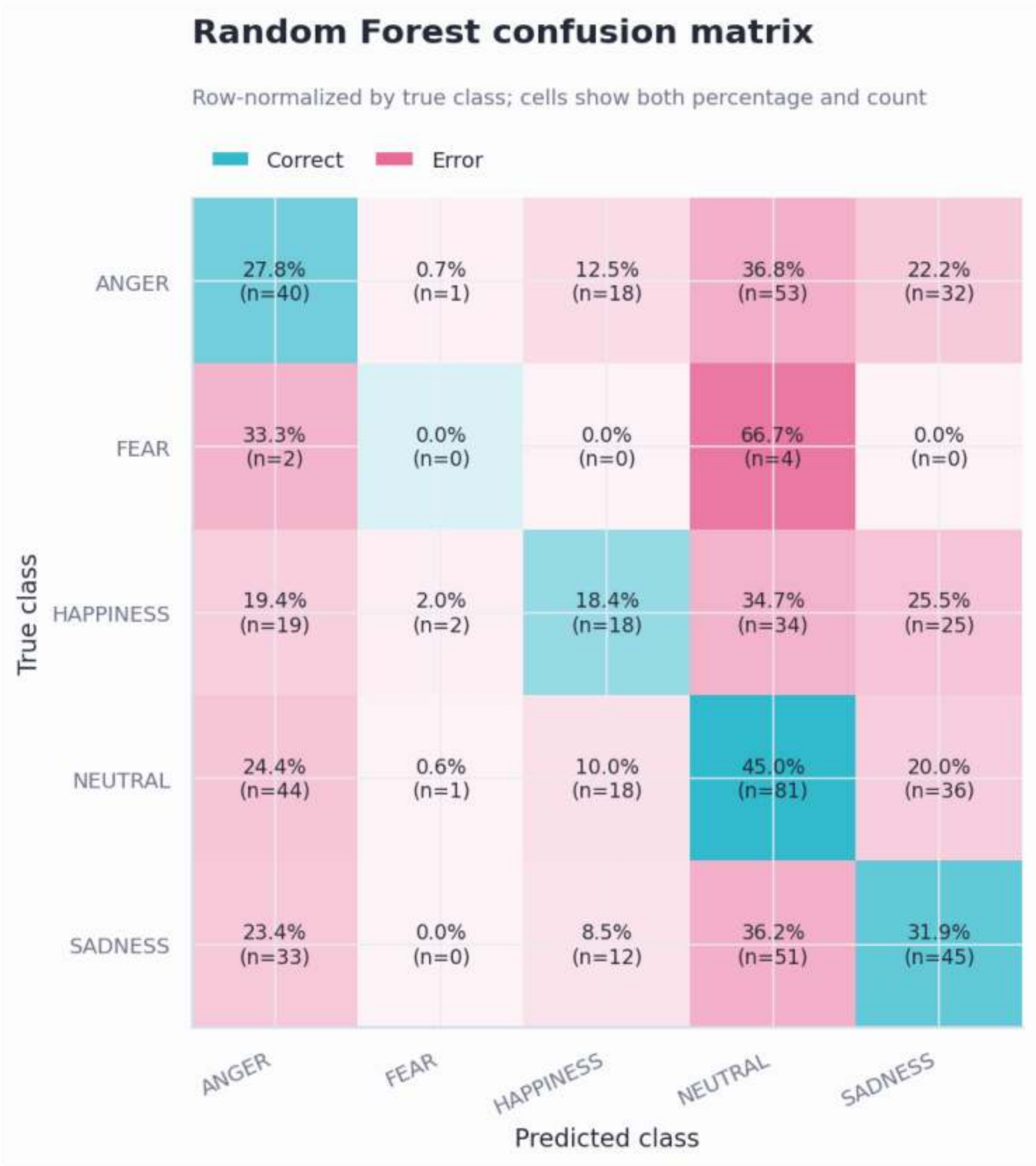
Identical 569-sample test split; FEAR has only 6 observations

Accuracy 0.323 · Kappa 0.079 · Macro F1 0.246

	Precision	Recall	F1	Support
ANGER	0.290	0.278	0.284	144
FEAR	0.000	0.000	0.000	6
HAPPINESS	0.273	0.184	0.220	98
NEUTRAL	0.363	0.450	0.402	180
SADNESS	0.326	0.319	0.323	141
macro avg	0.250	0.246	0.246	569
weighted avg	0.316	0.323	0.317	569

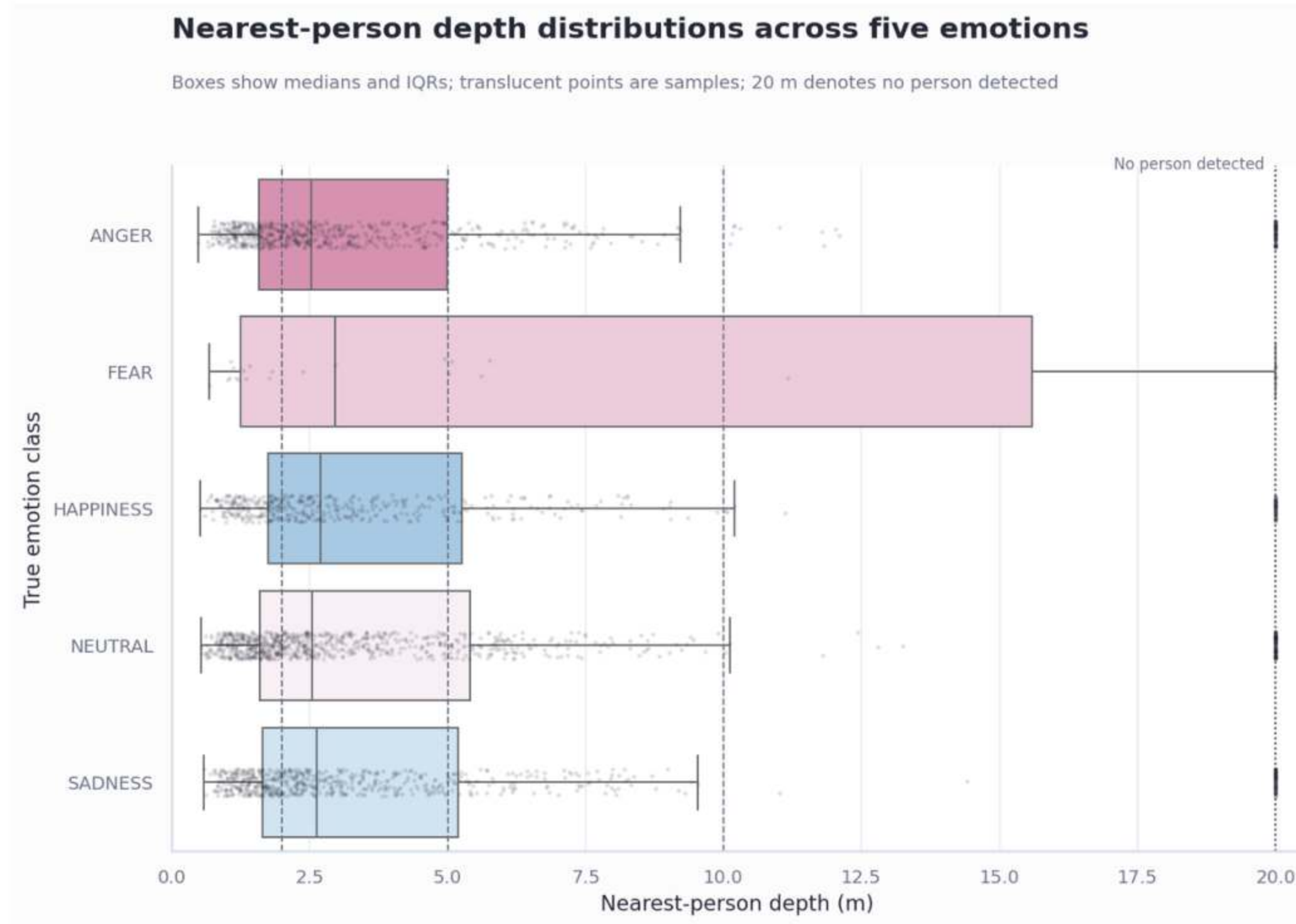
RF:Exploring Streetscape Contributions to Emotion Prediction

Classification Confusion and Pedestrian-Distance Effects



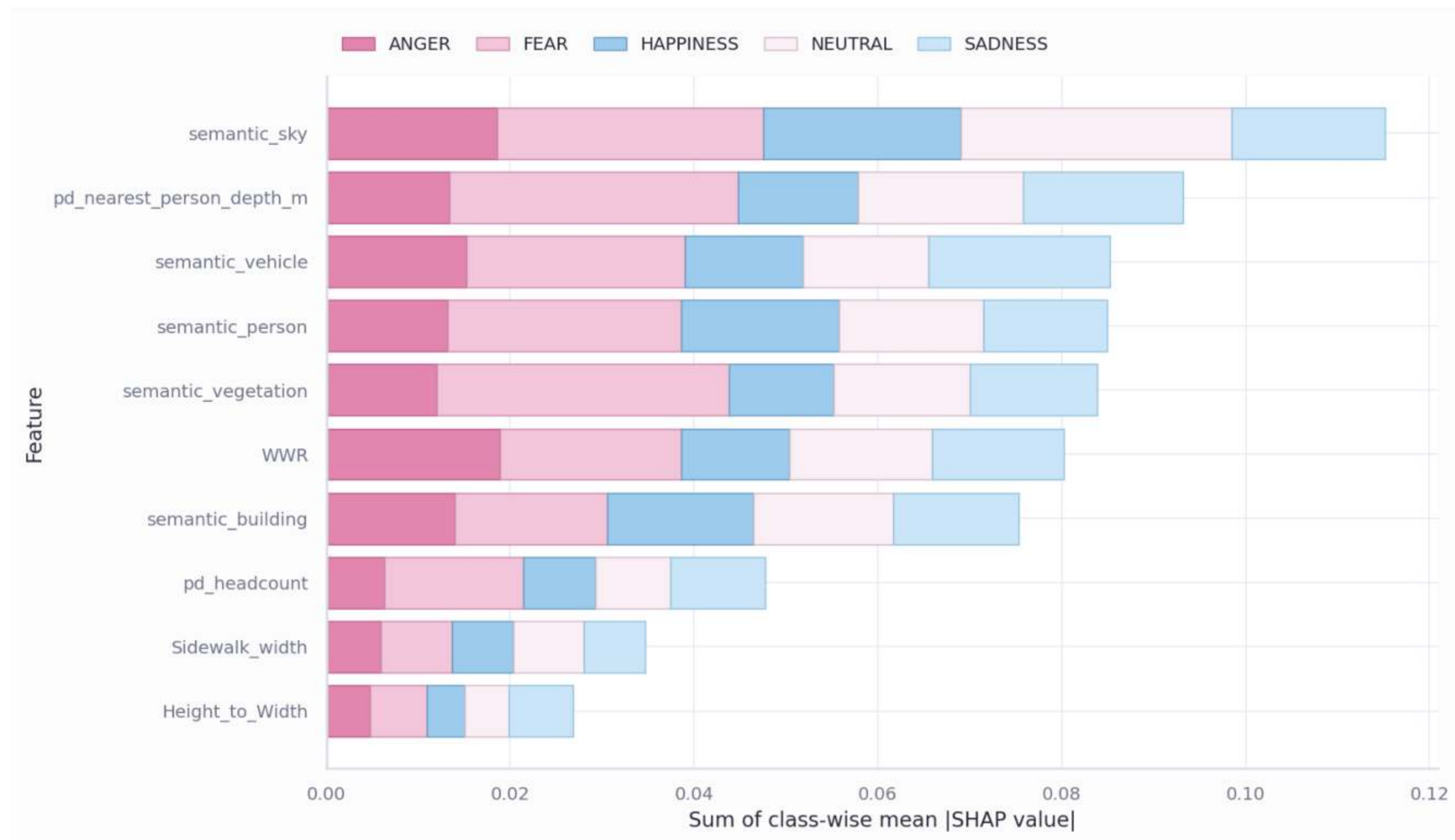
RF:Model Performance

Nearest-Person Distance by Emotion Class



SHAP:Random Forest global SHAP feature importance

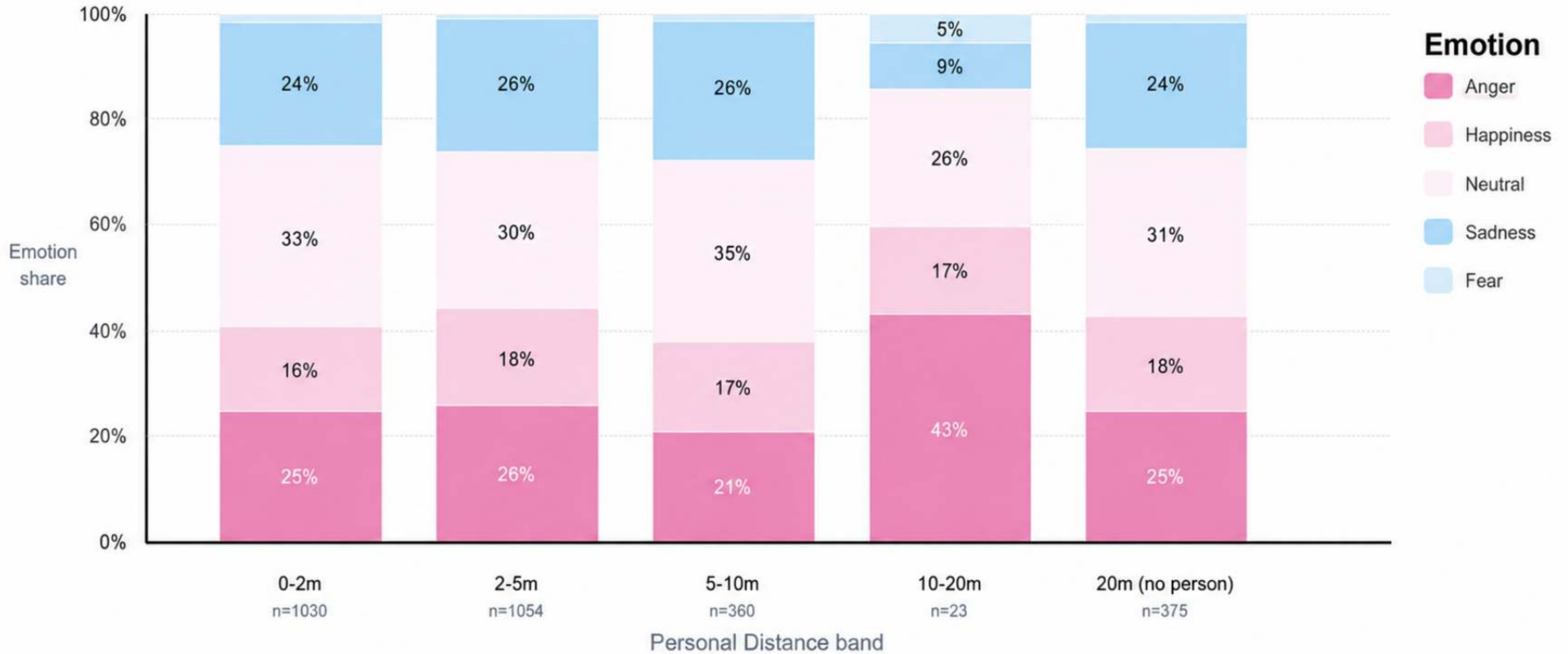
Identical 569-sample test split; colors represent the five emotion classes



Personal Distance ranks near the top of the SHAP contribution list, making it a corepeople-related spatial cue rather than a minor control variable.

Personal Distance: Emotion Composition Across Personal Distance Bands

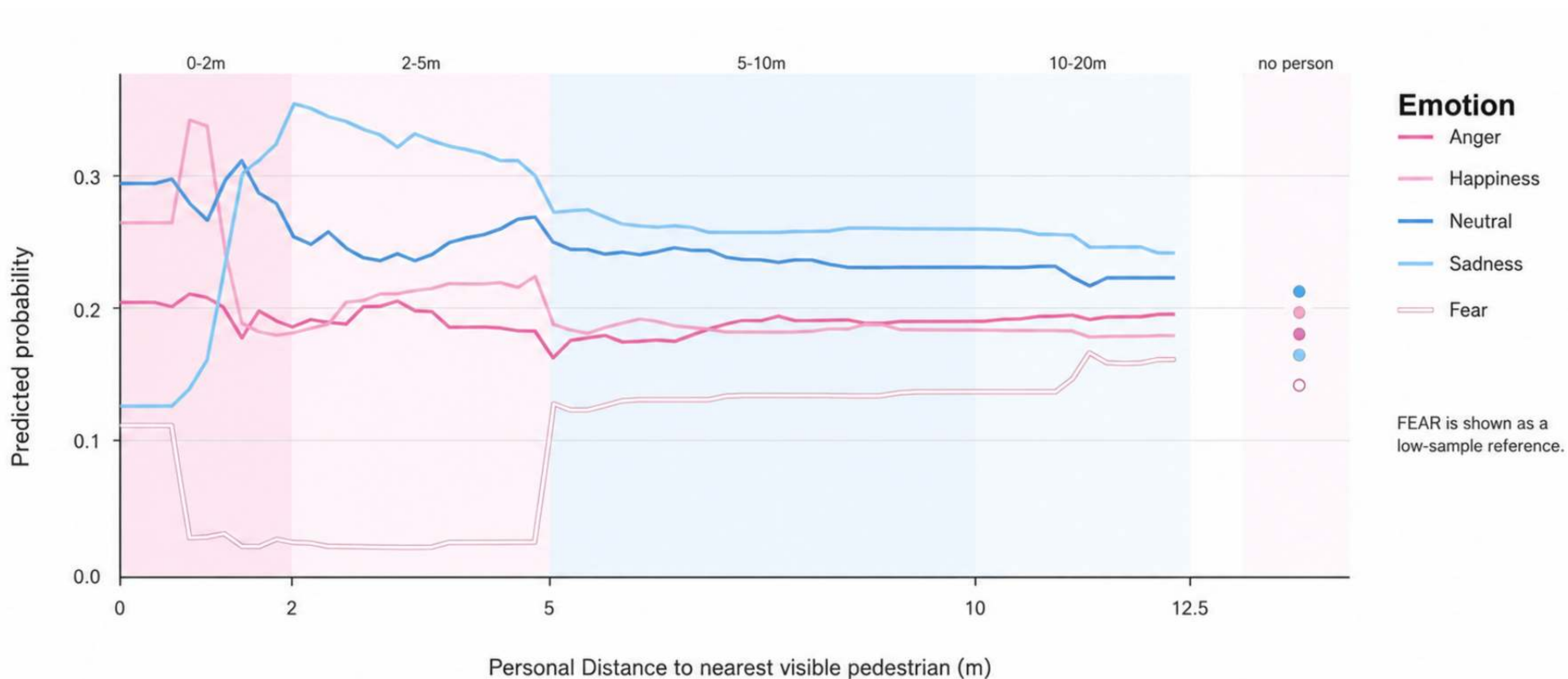
Stacked bars show how predicted emotional states vary as nearest pedestrian distance changes



The overall emotional composition remains relatively stable across different interpersonal distance ranges. However, in the 10 – 20 m range, the proportion of anger increases noticeably, while the proportion of sadness decreases. This suggests that Personal Distance may influence the emotional composition within a specific distance range.

Personal Distance:Effect of Personal Distance on Emotion Probability

Partial dependence: other spatial indicators are held at their average values



Changes in Personal Distance alter the predicted probabilities of the five emotion categories in a nonlinear manner. The fluctuations are most pronounced at close distances, while as the distance increases, the predicted probabilities of all emotion categories gradually become stable.

Personal Distance x Pedestrian Pixel Ratio

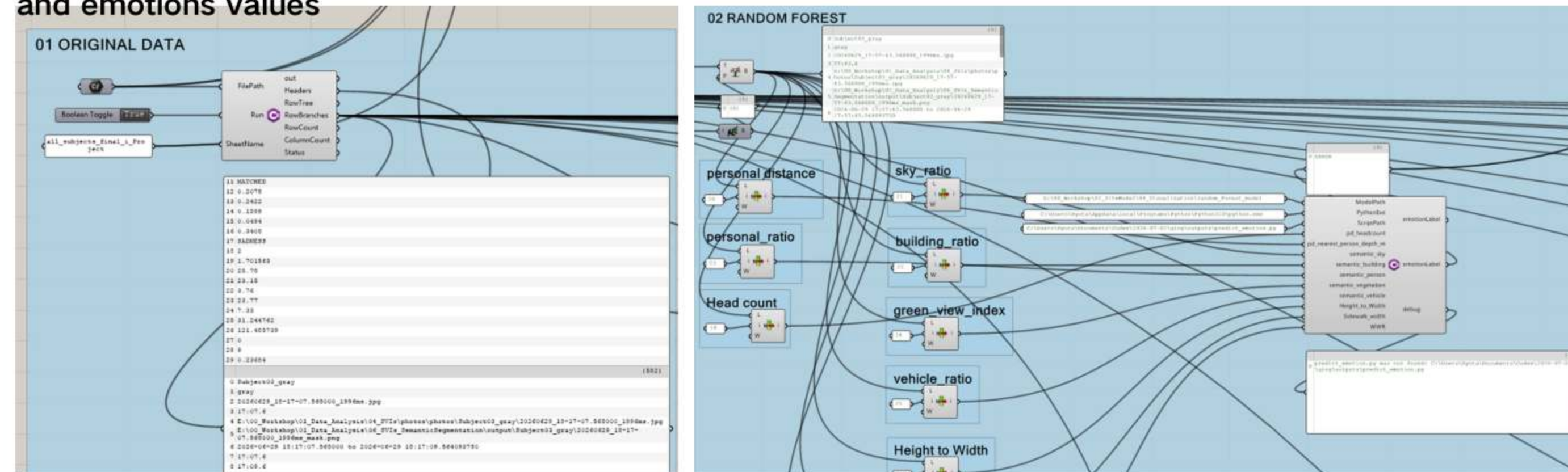
Cell color shows average Happiness probability; labels include sample size



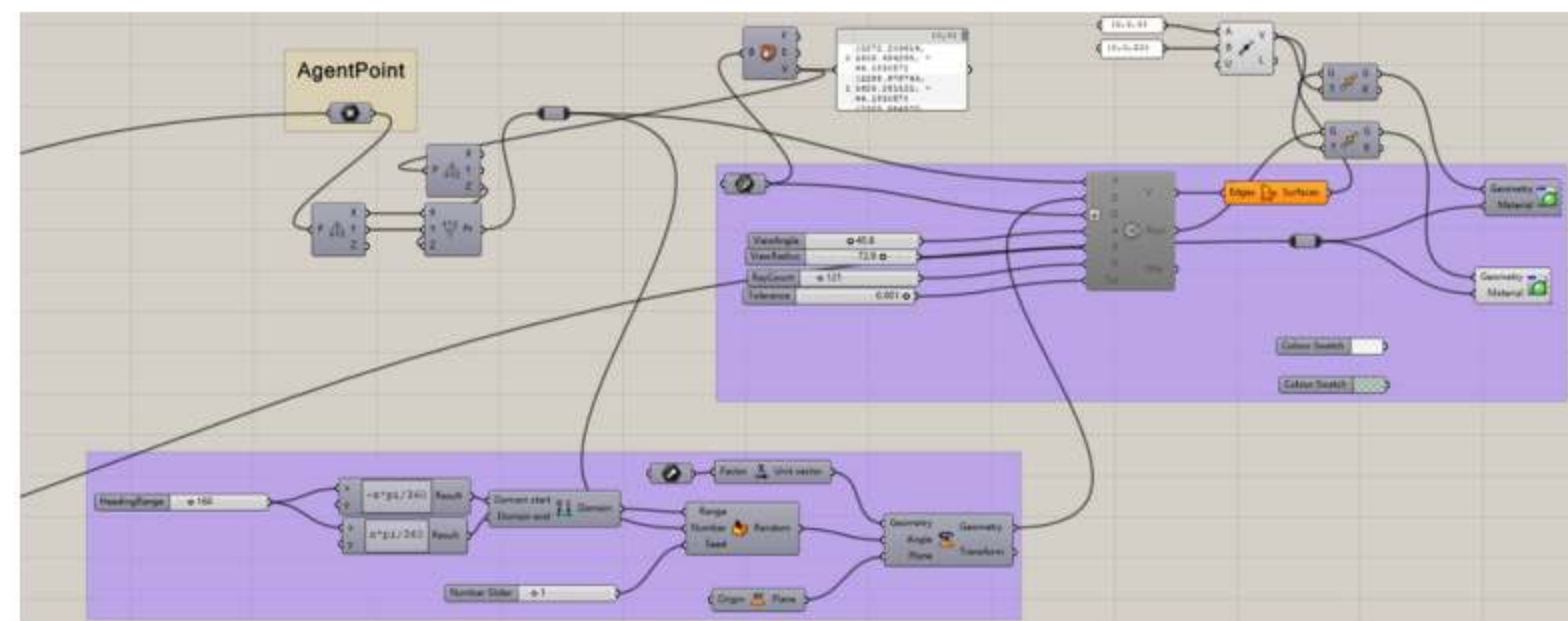
Pleasant emotions are more likely to occur under the combination of moderate pedestrian visibility and near-to-medium interpersonal distance. When the proportion of pedestrians in the image is too high, or when there are no clearly visible pedestrians nearby, the probability of Happiness instead decreases.

Construct the customized framework

Real-world track data with spatial indicators and emotions values	Pre-Trained RF model emotion prediction
	



Simulate human pedestrians walking with passers-by



The route becomes an emotional sequence, not a static map



We make spatial and interpersonal experience visible — and designable.

01

Measure

Synchronize EEG and location with environmental, enclosure and human indicators.

02

Interpret

Compare spatial contributions, then examine Personal Distance in depth.

03

Design

Return emotional evidence to Rhino for scenario comparison and spatial decision-making.

Next step

Validate with more participants and compare alternative design scenarios.